

8 Wi-Fi

The `wifiManager` represents the Android Wi-Fi Connectivity Service. It can be used to configure Wi-Fi network connections, manage the current Wi-Fi connection, scan for access points and monitor changes in Wi-Fi connectivity.

As with the Connectivity Manager, you access the Wi-Fi Manager using the `getSystemService` method, passing in the `Context.WIFI_SERVICE` constant-

```
String service = Context.WIFI_SERVICE;  
WifiManager wifi = (WifiManager) getSystemService(service);
```

To use the Wi-Fi Manager, your application must have `uses-permissions` for accessing and changing the Wi-Fi state included in its manifest.

```
<uses-permission android:name="android.permission.ACCESS_WIFI_STATE"/>  
<uses-permission android:name="android.permission.CHANGE_WIFI_STATE"/>
```

You can use the Wi-Fi Manager to enable or disable your Wi-Fi hardware using the `setWifiEnabled` method, or request the current Wi-Fi state using the `getWifiState` or `isWifiEnabled` methods:

```
if (!wifi.isWifiEnabled())  
    if (wifi.getWifiState() != WifiManager.WIFI_STATE_ENABLING)  
        wifi.setWifiEnabled(true);
```

The `WifiManager` also provides low-level access to the Wi-Fi network configurations. You have full control over each Wi-Fi configuration setting, which enables you to completely replace the native Wi-Fi management application if required. Later in this section you'll get a brief introduction to the APIs used to create, delete and modify network configurations.

8.1 Monitoring Wi-Fi Connectivity

The Wi-Fi Manager broadcasts `Intents` whenever the connectivity status of the Wi-Fi network changes, using an action from one of the following constants defined in the `WifiManager` class:

- **WIFI_STATE_CHANGED_ACTION:** Indicates that the Wi-Fi hardware status has changed, moving between enabling, enabled, disabling, disabled and unknown. It includes two extra values keyed on `EXTRA_WIFI_STATE` and `EXTRA_PREVIOUS_STATE` that provide the new and previous Wi-Fi states, respectively.
- **SUPPLICANT_CONNECTION_CHANGE_ACTION:** This `Intent` is broadcast whenever the connection state with the active supplicant